

Homework 4

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Problem 1:

(1) Using the formula

$$\pi_m(X)f = -\frac{\partial f}{\partial z_1}(X_{11}z_1 + X_{12}z_2) - \frac{\partial f}{\partial z_2}(X_{21}z_1 + X_{22}z_2),$$

by direct calculation we can write out the three Lie algebra generators in the representation **4** as

$$\pi_3(J_1) = \mathbf{i} \begin{pmatrix} 0 & -1 & 0 & 0 \\ -3 & 0 & -2 & 0 \\ 0 & -2 & 0 & -3 \\ 0 & 0 & -1 & 0 \end{pmatrix},$$

$$\pi_3(J_2) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -3 & 0 & 2 & 0 \\ 0 & -2 & 0 & 3 \\ 0 & 0 & -1 & 0 \end{pmatrix},$$

$$\pi_3(J_3) = \mathbf{i} \begin{pmatrix} -3 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

(2) The dual representation for a Lie algebra is defined by $\pi^*(X) = -\pi(X)^T$. Since $\pi_3(X)$ and $\pi_3^*(X)$ are both irreducible representations of $\mathfrak{su}(2)$ with the same dimension 4, they must be isomorphic (see B.Hall p96, theorem 4.32).

Problem 2: The matrix form of J_i in the representation **3** is

$$\pi_2(J_1) = i \begin{pmatrix} 0 & -1 & 0 \\ -2 & 0 & -2 \\ 0 & -1 & 0 \end{pmatrix}, \quad \pi_2(J_2) = \begin{pmatrix} 0 & 1 & 0 \\ -2 & 0 & 2 \\ 0 & -1 & 0 \end{pmatrix}, \quad \pi_2(J_3) = i \begin{pmatrix} -2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

We make a change of basis to a new set defined as

$$u_1 = \frac{1}{2}(z_1^2 - z_2^2), \quad u_2 = \frac{i}{2}(z_1^2 + z_2^2), \quad u_3 = z_1 z_2,$$

and write out the matrix form of $L_k = \frac{1}{2}\pi_2(J_k)$, we find

$$L_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad L_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad L_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

which is the standard generators of $\mathfrak{so}(3)$.

Problem 3: The Cartan subalgebra elements are diagonal matrices

$$H = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_{n+1} \end{pmatrix},$$

with

$$\lambda_1 + \lambda_2 + \cdots + \lambda_{n+1} = 0.$$

Then for matrix $E_{j,k}(j, k = 1, 2, \dots, n+1)$, we can compute

$$[H, E_{j,k}] = (\lambda_j - \lambda_k)E_{j,k}.$$

Hence when $j \neq k$, $E_{j,k}$ is a root vector for the root called $\alpha_{j,k}$, which satisfy

$$\langle \alpha_{j,k}, H \rangle = \lambda_j - \lambda_k.$$

Take the orthonormal basis $e_1, e_2, \dots, e_{n+1}$ of the vector space $\mathbb{C}^{n+1}$ such that $e_i = E_{i,i}$, we can write

$$p \cdot \text{tr}(\alpha_{j,k}^\dagger H) = \lambda_j - \lambda_k \Rightarrow \alpha_{j,k} = \frac{e_j - e_k}{p}$$

with coroots

$$H_{j,k} \equiv H_{\alpha_{j,k}} = \frac{2\alpha_{j,k}}{\langle \alpha_{j,k}, \alpha_{j,k} \rangle} = e_j - e_k.$$

The corresponding $X_\alpha = E_{j,k}$ and $Y_\alpha = E_{k,j}$ together with H_α satisfy the commutation relation of $\mathfrak{sl}(2, \mathbb{C})$.

Problem 4:

(1) The Cartan subalgebra of $\mathfrak{sl}(2, \mathbb{C}) \oplus \mathfrak{sl}(2, \mathbb{C})$ is $\mathfrak{h} = \text{span}\{H_1, H_2\}$ where

$$H_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \oplus O, \quad H_2 = O \oplus \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Then we can choose $\alpha_1 = H_1, \alpha_2 = H_2$ such that all the roots are $\pm\alpha_1$ and $\pm\alpha_2$, so a standard choice of simple roots is $\Delta = \{\alpha_1, \alpha_2\}$.

(2) The inner product gives

$$\langle \alpha_i, \alpha_j \rangle = 2\delta_{ij},$$

so the roots have length $\sqrt{2}$. Hence we can write $\pm\alpha_1, \pm\alpha_2$ as vectors

$$\alpha_1 = (\sqrt{2}, 0), \quad \alpha_2 = (0, \sqrt{2}), \quad -\alpha_1 = (-\sqrt{2}, 0), \quad -\alpha_2 = (0, -\sqrt{2}).$$

(3) Two basic operations are

$$s_{\alpha_1} \cdot (x, y) = (-x, y), \quad s_{\alpha_2} \cdot (x, y) = (x, -y),$$

and these two operations commute. Thus the Weyl group must contain four action:

- (1) Identity: e ;
- (2) Reflection across y -axis: s_{α_1} ;
- (3) Reflection across x -axis: s_{α_2} ;
- (4) Rotation by 180° : $s_{\alpha_1}s_{\alpha_2}$.